

NAUBINWAY, MI, USA

WMO: 997793

Lat: 46.0869N

Lon: 85.4444W

Elev: 180

StdP: 99.18

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-21.6	-19.2	N/A	N/A	N/A	N/A	N/A	N/A	14.4	-5.5	12.4	-4.1	2.4	300	0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.8	24.4	N/A	23.3	N/A	22.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.7	11.0	9.3	DB	-26.0	27.7	3.9	1.5	-28.9	28.7	-31.2	29.6	-33.4	30.4	-36.2	31.4	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.5	-8.1	-8.4	-3.5	3.2	9.6	14.9	17.9	18.0	14.7	8.6	2.2	-3.8	(6)	
	DBStd	10.35	5.91	5.59	5.61	3.77	3.56	3.06	2.74	2.48	3.38	3.91	4.55	4.85	(7)	
	HDD10.0	2493	559	516	419	205	50	2	0	0	3	74	234	428	(8)	
	HDD18.3	4746	818	749	678	454	272	109	41	37	116	302	483	687	(9)	
	CDD10.0	858	0	0	0	1	37	149	246	248	144	31	1	0	(10)	
	CDD18.3	68	0	0	0	0	1	6	29	26	7	0	0	0	(11)	
	CDH23.3	100	0	0	0	0	2	13	50	32	3	0	0	0	(12)	
	CDH26.7	6	0	0	0	0	0	1	4	1	0	0	0	0	(13)	
(14) Wind	WSAvg	4.2	4.4	4.2	4.2	4.6	4.2	3.6	3.3	3.6	4.2	4.9	4.9	4.5	(14)	
(15) Precipitation	PrecAvg	849	56	40	55	65	74	78	75	75	106	86	75	54	(15)	
	PrecMax	1132	116	95	173	143	123	178	127	128	215	163	114	129	(16)	
	PrecMin	609	25	11	3	20	18	30	29	28	25	34	30	3	(17)	
	PrecStd	121	23	21	37	34	27	34	30	28	43	38	23	28	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.6	3.2	12.1	13.8	22.1	25.3	26.9	25.7	23.7	19.2	12.8	7.6	(19)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)	
	2%	DB	2.3	1.6	8.2	11.7	18.7	22.6	24.6	24.1	21.8	17.5	10.9	5.6	(21)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)	
	5%	DB	1.4	0.7	5.5	10.1	16.7	20.7	23.5	23.1	20.9	16.1	9.7	4.2	(23)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)	
	10%	DB	0.4	-0.2	3.4	8.6	14.9	19.2	22.3	22.2	19.9	14.9	8.7	2.5	(25)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)	
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)	
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)	
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)	
(35) Mean Daily Temperature Range	MDBR		7.0	8.1	8.2	6.5	7.0	6.8	6.6	6.8	6.7	6.0	5.3	5.7	(35)	
	5% DB	MCDBR	5.2	6.5	8.0	8.6	9.1	8.8	7.8	7.3	6.7	6.4	5.4	6.2	(36)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)	
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.283	0.288	0.313	0.354	0.376	0.393	0.391	0.380	0.367	0.335	0.303	0.275	(40)	
	taud		2.412	2.383	2.404	2.352	2.362	2.348	2.373	2.405	2.432	2.503	2.521	2.485	(41)	
	Ebn at Noon		825	894	919	903	890	874	871	867	848	829	799	802	(42)	
	Edh at Noon		70	88	100	116	119	121	117	109	97	79	64	59	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.10	2.03	3.37	4.52	5.60	6.06	6.09	5.16	3.88	2.26	1.19	0.84	(44)	
	RadStd		0.12	0.21	0.37	0.49	0.46	0.42	0.29	0.33	0.30	0.26	0.16	0.08	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page